The Fable Computer, Part II: Quantum-Limited Analog Tensor Processing on the Regenerative Graphene-Plasmon Fabric

Interference computing with mesoscopic plasmon pulses, classical on-chip decoding to 1–2 bits, and the temperature budget of the quantum–classical crossover

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Authorship and Status Disclaimer. This manuscript extends the author's design study [1] ("Part I" throughout). Generation of content involved the use of an AI, Anthropic's Fable 5, at the author's direction, including construction and adversarial cross-checking of the released model package (Appendix QA). The account of the community notes promoted from 31 July through 17 October 2026 was prepared in a maintainer-operated Claude Code session in which Claude Fable 5.1 orchestrated Claude Opus 5.5 agents for mapping, two-lens adversarial verification, drafting and checking; the community notes the text relies on were themselves produced and assessed by the project's disclosed agent pipeline, each note's header giving its promotion route and, where the note records them, its executing models. The author declares a lack of formal expertise in quantum optics and graphene-plasmonic device physics, and the document has not received independent expert review. Model claims are reduced-order results in the sense of Part I ("in-model": established within a named model — the released chain or a stated extension of it — under its stated assumptions, not measured), and must be treated as a starting point for discussion, not as peer-reviewed engineering guidance. Statements carry the labels of the community record: "demonstrated" means executed on the released, unedited model chain at the stated convention, or read or computed directly from its source (marked "source read" or "arithmetic") — a model result, not a measurement; "open" marks what neither the chain nor a measurement settles; and "external" marks facts from the literature.

Abstract

Part I designed a room-temperature terahertz half adder on a regenerative graphene-plasmon fabric and answered its feasibility question in-model. This Part asks what the same fabric computes when its pulses are treated as what they are: quantum fields. Quantizing the Part-I Dyakonov–Shur (DS) cell mode puts one plasmon at a fractional density amplitude of 0.16 %, the 1-dB compression knee at ≈38 plasmons and the coded logic rail at $\approx 3.8\times10^{3}$ — logic pulses are mesoscopic, and quantum noise is a quantitative design input, not a curiosity. That ladder is mode-energy quantization applied to a classical Part-I solver amplitude (the knee) and a coded rail constant whose swing no run in the merged community record (the promoted notes and the released code) delivers to a deciding input (Section 3.1); it fixes the mesoscopic framing and sizes the machine, and — as that record establishes — it is not what the no-go below rests on. Two results follow. First, an honest no-go: the single-plasmon Kerr nonlinearity of the 576-nm cell falls short of its loss rate by roughly five orders of magnitude ($\chi/\kappa \approx 8.2\times10^{-6}$ at 4 K at the carried Kerr coefficient; ≈4.5–5.8 orders at 4 K across its open c_K = 0.05–1 band), so gate-model (photon-blockade) quantum logic is excluded on this fabric at any temperature; the blockade would demand ~3-nm cells, consistent with the quantum-plasmonics literature. Second, what survives is exactly the machine the fabric's own architecture suggests: a quantum-limited analog (continuous-variable) computer, in which comb-locked coherent pulses interfere at gate-programmed junctions — plasmon–plasmon interaction in the analog-computer sense — and the result is quantized on-chip by the Part-I cells themselves, threshold cells by design, re-programmed as a 1–2-bit flash decoder (its two ladder comparators redesigned for the k ≥ 16

sharpness that decoding needs, and its inverter and AND carried over from Part I, though in-model the released cell supplies neither as a threshold gate — Section 5, Table QB1). The minimal tensor unit, QMAC-1 (two inputs, one interference, one output decoded to 2 bits), is specified to cell-map level; in its digital special case the decoded word is the binary count A+B, so the Part-I half adder re-emerges as the decoder of a single quantum interference. Error rates versus temperature are computed on one axis for the quantum-analog unit and the classical digital fabric: with a quiet launch and the decoder amplifier's idler (its noise input) at the lattice occupation (Sections 4.2 and 6), the 2-bit decode is unusable per-shot at 300 K (0.14), reaches 3.1×10^{-3} at 77 K, and saturates at a vacuum-set floor of 1.3×10^{-6} below ≈20 K (N = 400 plasmons per input); the 1-bit decode is already usable warm (3.9×10^{-3} at 300 K, conditional on the warm DS threshold that Section 1 leaves open) — at the design pulse energy, lower-bit decoding is what the physics supports per shot (Section 5). $T_Q = \hbar\omega_0/k_B \approx 48$ K is where the noise budget crosses from thermal- to vacuum-dominated; below ≈15–20 K the error curves saturate at vacuum-set floors that no further cooling improves. That saturation is the quantum limit made visible, and its observation is a pre-registered bench gate. Part I's no-cryogenics lock is explicitly relaxed here and the fork is stated as such.

1. Introduction: from restored bits to quantum-limited amplitudes

Part I [1] built a machine out of one physical object — a resonant DS cell [2] biased below its self-oscillation threshold — and one discipline: every gate restores its logic levels against a rail, so decisions never erode. The price of that discipline is nonlinearity: digital restoration lives at the saturating rail. (In the released model the level is re-clamped, but not at a rail: the cell as released has no saturating output plateau, demonstrated, and in-model a loaded fan-out-2 cascade, composed from a single-cell sweep with no multi-cell run, re-clamps near a 2 % drive — the cell compresses, it does not threshold; Sections 3.1 and 5.) This Part starts from the opposite end of the same transfer curve. Below the 1-dB knee the fabric is a linear, phase-coherent wave medium: pulses split, merge, and interfere at gate-programmed junctions, and the comb-locked clock of Part I — distributed at femtosecond-class phase — gives every cell a common phase reference. A linear interferometer with programmable weights is an analog computer: it multiplies by transmission amplitudes and adds by superposition. The question Part II asks is how well that analog computer can compute, and the answer is a quantum one, because the pulses in play are small.

Quantizing the Part-I operating point (Section 3) puts the knee of a cell at ≈38 plasmon quanta and the coded rail at ≈3833 — mode-energy bookkeeping applied to a classical solver amplitude and a coded rail constant (Section 3.1), which is what makes the framing mesoscopic and what sizes the machine, but which is not, and is not used as, an argument about quantum logic (Section 3.3). At those occupations three quantum facts dominate the machine design. The 1-THz quantum is small against the room-temperature bath ($\hbar\omega_0/k_B \approx 48$ K), so at 300 K every mode carries ≈5.8 thermal plasmons of noise; below ≈48 K vacuum fluctuations take over — the exact crossover, where the thermal occupation falls to the vacuum half-quantum, is $T^* = \hbar\omega_0/(k_B \ln 3) \approx 43.7$ K (promoted community derivation, July 2026) — and by ≈15–20 K the noise floor stops improving altogether. Phase-insensitive gain — the only kind the DS cell provides under its DC bias [2, 3] (a $2f_0$-pumped cell is phase-sensitive in narrow windows, in-model; Section 7) — adds at least the Caves quantum noise. Referred to a matched thermal input,

the released gain rule's noise figure (Section 4.1) is the $F = 2 - 1/G$ floor Part I already carried as its Eq. (7); its absolute added noise, $(G - 1)(\bar{n} + ½)$, is $2\bar{n} + 1$ times the Caves minimum (×12.5 at 300 K, ×14.7 at 353 K), and the two coincide only as $T \to 0$ (in-model; notes/2026-08-05-idler-added-noise-and-the-nf4db-requirement.md §5 — the dated notes cited by file path throughout this Part are the promoted notes of the community record, one Markdown file each in the notes/ directory of the repository, https://github.com/ryoji-info/FableComputer/tree/main/notes, where notes/INDEX.md lists what each settles and what later notes did to it, and each note's header gives its promotion route). And amplitude resolution per shot is bounded by shot noise at a given pulse energy, so the number of bits one pulse can carry is finite and computable (Section 5). The machine that emerges is not a quantum computer in the gate-model sense — Section 3.3 quantifies why that is excluded on this cell — but a quantum-limited analog tensor engine with a classical digital wrapper: the same fabric, the same five-cell block as Part I, re-programmed so that two of its cells act as comparators of a flash analog-to-digital ladder — those two redesigned for the $k \geq 16$ sharpness that decoding needs, and the inverter and AND carried over from Part I under a qualification of their own (Section 5).

One constraint of Part I is explicitly relaxed: no-cryogenics. The quantum-analog unit works at 300 K only in its 1-bit mode (16-slot averaging is a cold-class instrument, Section 5); its 2-bit per-shot mode wants ≤77 K with a quiet launch and the decoder amplifier's idler at the lattice occupation, and its quantum floor lives below ≈20 K (Section 6). Part II therefore defines temperature classes rather than one operating point, keeps the ≤80 °C cap for the warm class, and states plainly that the cold classes are a fork from Part I's charter, not a refinement of it. Cooling also changes the classical fabric itself (Part I, Section 8.3): the DS threshold tracks temperature, so all cold-side operation inherits Part I's tracked-bias requirement, and below ≈255 K the hydrodynamic closure has not been evaluated, so every cold-side gain figure here is uncertified and carries that caveat to the Boltzmann–Maxwell tier — a smooth signed penalty that deepens on cooling, not a hard expiry (Part I, Section 8.3). That penalty is a statement within the closure of notes/2026-07-13-kinetic-correction-signed-band.md, which keeps a DS threshold at the warm end. The warm class inherits an open question of its own: in the finite-temperature single-rate kinetic operator that note named as the next tier's decisive computation, no DS threshold exists below the drift's Landau ceiling (the drift at which the slow cavity leg's phase speed falls to graphene's v_F) at 353 K for $A \leq 1$ (A the prefactor of the electron–electron rate, not the mode area of Eq. (Q1); run at 0.5, 1 and 3), and at 300 K one exists only at $A = 3$ with the electrons shedding their drift heat at once ($\tau_E \to 0$) — in-model, on the runs whose driven base state is stable; whether graphene's own electron fluid has a threshold at 353 K is open, since three admissible two-rate continuations give three different answers (notes/2026-10-15-graphene-fluid-threshold-turns-on-current-channel.md). Every warm gain figure in this Part, the 1-bit decode and QG4's model target included, is therefore conditional on the cell having a DS threshold at its operating temperature, which at 300 K no two-rate continuation has tested.

2. Related work and positioning

Quantum plasmonics. That surface plasmons preserve quantum coherence is established: entangled photons survive conversion through plasmonic channels [4], and the field has a developed single-quantum toolbox [5]. For graphene specifically, the extreme mode

confinement prompted the prediction that few-nanometre graphene nanostructures could reach single-plasmon Kerr nonlinearities of order unity [6] — the photon-blockade regime. Section 3.3 places the Part-I cell against exactly that scaling and finds it roughly five orders of magnitude away (≈4.5–6.1 orders across the open c_K band at the two temperatures Section 3.3 quotes): this design claims the opposite corner of quantum plasmonics, many-quantum Gaussian operation at the standard quantum limit.

Continuous-variable computing. Encoding analog values in field quadratures and processing them with Gaussian operations is the continuous-variable framework [7, 8]; universal quantum computation in that framework is known to require a non-Gaussian resource [7], which is precisely what the Kerr no-go denies this fabric. What Gaussian operations do provide is quantum-limited analog arithmetic: the amplifier bound of Caves [9] (anticipated by Haus and Mullen [10]) sets the noise cost of gain, and homodyne-class detection against vacuum noise sets the per-shot precision at a given pulse energy.

Photonic analog accelerators. Coherent interferometric multiply-accumulate is the operating principle of photonic neural accelerators, from programmable nanophotonic processors [11] to comb-based convolutional tensor cores [12]. The niche claimed here is different in three ways: the carrier is a THz plasmon confined ≈100× below the free-space wavelength (footprint), the weights and routing are gate-voltage-programmable on one chip, and — the load-bearing difference — the analog result is quantized, by design, by the same regenerative cell family that computes, so no electronic ADC enters the signal path (Section 5, which qualifies the decoder cells). The cost is the THz quantum: at 1 THz the thermal bath is hot in quantum units at room temperature, which is why Section 6's temperature budget is the central result of this Part.

3. Quantum scales of the Part-I operating point

3.1 How many quanta is a logic pulse?

The hydrodynamic energy of the acoustic (gate-screened) standing mode [13] of one DS cell (length L ≈ 576 nm, effective area A ≈ 10^{-8} cm^2, density n = 10^{12} cm^{-2}, speed s from Part-I Eq. (1)) with fractional density amplitude ε = δn/n is

$$E(\varepsilon) = \tfrac{1}{4}\, n\, m^* s^2 \varepsilon^2 A, \quad m^* = E_F / v_F^2 \qquad \text{(Q1)}$$

(kinetic plus potential of the shallow-water mode, spatially averaged over the standing wave). Setting E = $\hbar\omega_0$ defines the single-plasmon amplitude and the quantum content of any classical swing:

$$N(\varepsilon) = (\varepsilon/\varepsilon_1)^2, \quad \varepsilon_1 = \sqrt{(4\hbar\omega_0 / n\, m^* s^2 A)} \approx 1.62\times10^{-3} \qquad \text{(Q2)}$$

One plasmon is a 0.16 % density swing. The Part-I solver's 1-dB gain-compression knee (ε ≈ 1 %) is therefore N ≈ 38 quanta (0.025 aJ; input-/launch-referred, band 22–51 across the CW and pulse observables on the solver's N = 240 grid with the bias at 0.7 of that grid's own threshold, where N counts grid cells, not quanta — the intracavity reading is a reference-plane mislabel), and the ≈10 % rail is N ≈ 3833 (2.5 aJ) — a figure anchored at neither reference plane: its intracavity reading is contradicted by the released solver's own compression (−0.35 dB at a 10.05 % intracavity swing, against the −14.15 dB that the coded compression form cell.compression_dB assigns to a 10 % swing) and its drive-plane reading is a gain-compression

point, not an output level. On the released solver at 353 K, in the retuned cavity configuration that notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md runs (operating cell length 576.62 nm, not the shipped zero-drift 582.80 nm), a 10 % source drive is compressed by 9.1–13.7 dB on the N = 240–720 grids on the absolute and ratio bias legs, depending on leg and observable — a bias leg being where the drift bias is held as the grid is refined: at its N = 240 value on every grid (absolute), at 0.7 of each grid's own threshold (ratio), or at 0.7 of the analytic threshold (analytic). That −14.15 dB is reached only beyond these grids, on the absolute leg's CW observable, between the N = 960 grid (−14.03 dB, where that fixed bias sits at 0.77 of the grid's own threshold) and the N = 1200 grid (−14.25 dB). Without a parametric pump (Section 7) the cell's output keeps growing on those two legs as a power 0.39–0.60 of the drive, from a 3 % to a 70 % drive, with no plateau. These are demonstrated for the released solver — a shallow-water model with Galilean fluid kinematics and no velocity cap, whose fluid speed exceeds graphene's v_F above a ≈3 % drive — so they are statements about the model, not the physical cell. In-model, a loaded fan-out-2 cascade re-clamps near a 2 % drive (99–277 quanta on the absolute and ratio legs, 43–221 on the analytic leg): a fixed point read off this single-cell sweep under the cascade construction of notes/2026-08-01-gated-requirement-round-trip.md, which takes a source-driven single cell's gain as the cell-to-cell gain (no multi-cell cascade was run); under that note's gated duty-0.8 launch, run in the shipped configuration, the loaded fixed point sits lower, at ≈75 quanta, band 27–147. No run of the record delivers the 3833-quantum rail to a deciding input, and the rail's plane itself stays open. The knee band is likewise bound to grid and bias leg: on the N = 480 grid it is 21–43 quanta on the ratio leg, essentially unchanged, and 13–26 on the absolute leg, whose bias relative to the grid's own threshold drifts with refinement; on the analytic leg it is 57–122 on the N = 240 grid (demonstrated brackets, interpolated crossings; all from notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md). The provenance of the knee and rail values should be stated exactly: the knee ε is a classical Part-I shallow-water-solver output (the compression knee; promoted community derivation, July 2026), and the rail ε is a coded constant (qmode.EPS_RAIL) that notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md (Table 1) marks assumed, never produced by any run; both are imported here and converted to quanta by Eqs. (Q1–Q2). They are quantization bookkeeping applied to classical inputs, not outputs of the quantum noise chain, which is exactly linear and amplitude-scale-free in N and can generate no intrinsic occupation threshold of its own (Section 4.1). The arithmetic reproduces from Eqs. (Q1–Q2) against an independent hand reconstruction to ≈1.2 % in ε_1 and, because $N \propto \varepsilon_1^{-2}$, to ≈2.4 % in N_knee and N_rail (Appendix QA); the absolute calibration of Eq. (Q1) — the effective area A and the ¼ mode-energy prefactor, one hydrodynamic averaging convention — is owned by the Boltzmann–Maxwell tier (Section 8) and is open: a factor of two in the prefactor moves both N_knee and N_rail by two. Two scales should not be confused. Part I's “intrinsic field energy of ≈10 aJ per bit” counts a traveling few-cycle pulse over its ≈3λ_p in-flight extent (an area an order of magnitude larger than one cell); Eqs. (Q1–Q2) count quanta in the quarter-wave cavity mode of one cell, which is the mode the noise physics and the comparator decisions live in. The two are consistent — a traveling pulse at rail-class swing over 3λ_p carries 10–30 aJ — and both sit below the launch-side slot energy before the still-unquantified coupling (Part I, Section 9.5). By design, a logic pulse on this fabric is a mesoscopic object — tens to thousands of quanta — and Figure Q1(A) shows the whole ladder. What that ladder anchors is the mesoscopic framing of this Part and the classical machine design that follows from it — the passive-core rule of Section 4.2 and the decoder

sizing of Section 5. It does not anchor the no-go of Section 3.3, whose logic runs on entirely different quantities; the two roles are kept separate throughout.

3.2 The bath: 48 K in quantum units

The 1-THz quantum is $\hbar\omega_0 \approx 4.14$ meV, so the crossover temperature is $T_Q = \hbar\omega_0/k_B \approx 48$ K. Table Q1 gives the Bose–Einstein occupation of the mode across the temperatures used in this Part. At the Part-I band (300–353 K) every mode carries six to seven thermal quanta — the fabric computes while swimming in a bath an order of magnitude above the vacuum floor; at 77 K one thermal quantum remains; at 20 K the mode is effectively dark. The plasmon lifetime improves in parallel but saturates: $\tau(T)$ follows Part I's $1/T$ phonon scaling down to ≈150 K and is clamped by impurity scattering below (Part I, Section 8.3), so Q saturates at ≈12.6. The saturated $\tau = 2$ ps sits slightly above the best measured cryogenic graphene-plasmon lifetimes (≈1.6 ps [14], an infrared near-field imaging measurement, external; the THz-band lifetimes cited in Part I, Section 4, are room-temperature ones, so the comparison crosses bands) — a premium-material assumption in exactly the sense of Part I's $\tau(300\text{ K}) = 1$ ps, and flagged as such; whether scalable graphene/hBN stacks reach that 1 ps is an open design question (issue #140). Cooling buys a quieter bath much faster than it buys a better resonator.

T (K)	$\bar{n}$ (thermal quanta)	$Q = \omega\tau$	per-gate loss (dB, $\lambda_p/2$)
353	6.87	5.3	2.56
300	5.76	6.3	2.17
150	2.65	12.6	1.09
77	1.16	12.6 (sat.)	1.09
48 (= T_Q)	0.58	12.6 (sat.)	1.09
20	0.10	12.6 (sat.)	1.09
4	6.2×10^{-6}	12.6 (sat.)	1.09

Table Q1. The 1-THz mode against its bath. Lifetime saturation below ≈150 K follows Part I's stated cold-side caveat; premium (graphite-gated) material is carried as upside, not assumed.

3.3 The honest no-go: no photon blockade on this cell

Gate-model quantum logic needs one quantum to shift the cell's resonance by more than a linewidth — $\chi/\kappa \gtrsim 1$, with χ the single-plasmon Kerr rate and $\kappa = 1/\tau$ the decay rate. The hydrodynamic nonlinearity gives, at reduced order,

$$\chi \approx c_K\,\omega_0\,\varepsilon_1^{\,2}, \quad c_K = O(1)\ \text{band}\ 0.05\text{–}1\ (0.25\ \text{carried}) \qquad \text{(Q3)}$$

i.e. $\chi/2\pi \approx 0.7$ MHz against $\kappa/2\pi \approx 160$ GHz at 300 K: $\chi/\kappa \approx 4.1\times10^{-6}$ at 300 K and 8.2×10^{-6} even at the lifetime-saturated 4 K — five orders of magnitude short at the carried $c_K = 0.25$, and ≈4.5–6.1 orders once the open $c_K = 0.05$–1 band of Eq. (Q3) is taken in at both of the temperatures quoted here (≈4.5–5.8 orders at 4 K, ≈4.8–6.1 at 300 K) — short at any temperature this stack can reach, and nowhere by less than ≈4.5 orders, since the lifetime-saturated cold rows are the most favourable case the stack offers and warming only shortens τ and widens the gap. That band is the honest width of the "five orders" phrase (promoted community derivation, July 2026): the shortfall is robust at order-of-magnitude level, but its exponent is only as sharp as c_K, which the Boltzmann–Maxwell tier owns (Section 8). Because χ

$\propto 1/A$ at fixed density and frequency, closing the gap would demand ≈3 nm structures — a bookkeeping scale, not a device: no few-nanometre cell hosts a 1-THz quarter-wave mode at all (its resonance would sit near 200 THz), so blockade is excluded here twice over, by magnitude and by geometry. Both exclusions are consistent with the graphene quantum-plasmonics literature, whose blockade proposals live at few-nanometre scale and correspondingly mid-infrared frequencies [6]. The conclusion is stated plainly: this fabric supports no plasmonic qubit. Its load-bearing anchors are named as plainly: in-model the exclusion rests on the χ/κ shortfall, with its open c_K band, and on the geometry. The 38/3833 ladder of Section 3.1 is not among them — that ladder is quantization bookkeeping on a classical solver amplitude and a coded rail constant (Section 3.1) and plays no part in this exclusion (promoted community derivation, July 2026). Nor is the no-go fragile for resting on two things; it is over-determined. The released chain's own verified loss+thermal generator dephases a superposition of vacuum and an N-quantum pulse within a single gate segment for N of order unity at every temperature in this Part ($N^* \approx 0.4$ at 300 K, ≈9 at 4 K, per gate), which excludes a plasmonic qubit here more brutally than the Kerr gap does. That decoherence factor is a standard consequence of the Lindblad generator verified in Appendix QA and not a printed output of the released code; it is cited as a hand derivation, and its proximity to the knee at 4 K is a coincidence of two quantities with no shared parameter dependence, not a corroboration. The mesoscopic cross-Kerr interaction is real but sub-resolution per shot — a 400-quantum control pulse writes ≈6.6 mrad of phase in 2 ps, versus ≈25 mrad single-shot homodyne resolution at 4 K (≈15 shots to resolve): measurable as a systematic with averaging (bench gate QG5), not computational per slot. Everything that follows is therefore Gaussian: coherent states, linear optics, phase-insensitive gain, and homodyne-class decisions — quantum-limited analog computing, with the limits computed rather than assumed.

Figure Q1. Quantum scales of the plasmon fabric (operating point of Part I)

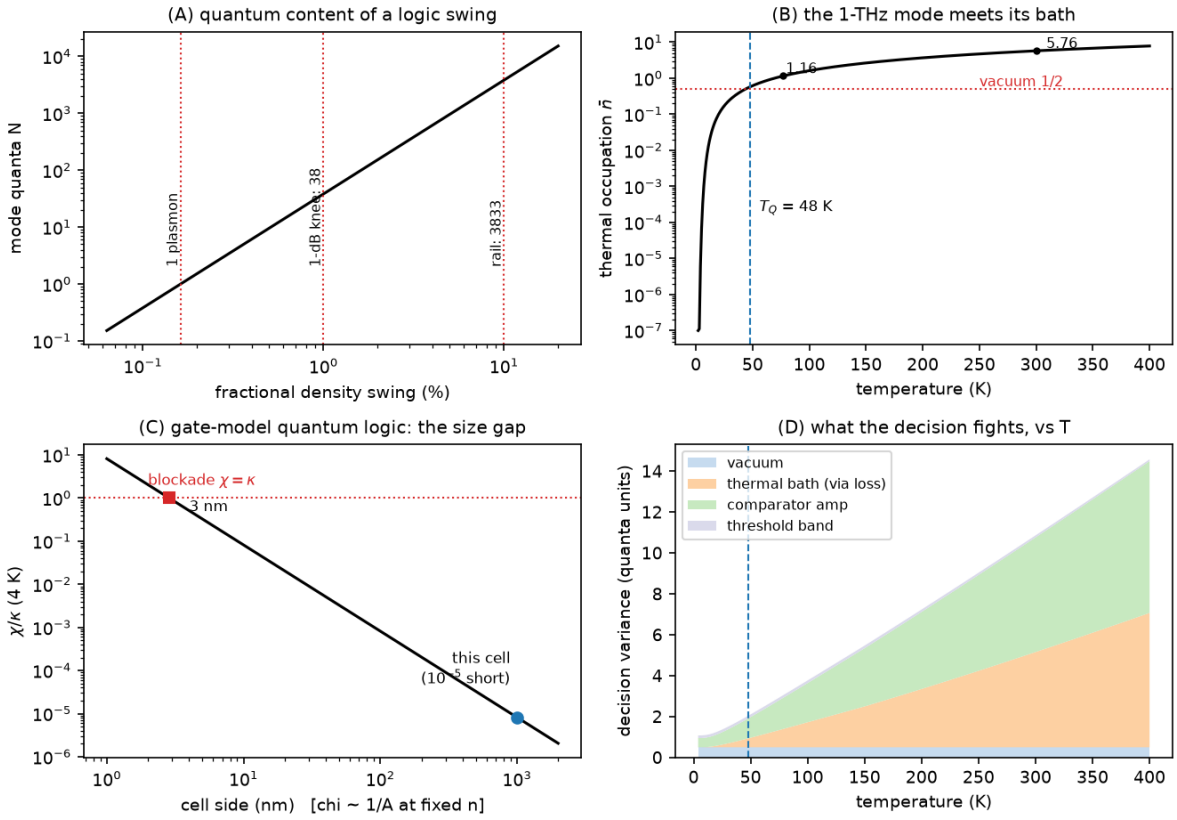


Figure Q1. Quantum scales of the fabric at the Part-I operating point. (A) Quanta in a classical swing, Eq. (Q2): one plasmon, the 38-quantum knee, the 3.8×10³-quantum coded rail. (B) Thermal occupation of the 1-THz mode: T_Q ≈ 48 K separates the thermal and vacuum regimes. (C) The blockade gap: χ/κ versus cell size at fixed density; the Part-I cell sits five orders below blockade at the carried c_K = 0.25 (≈4.5–5.8 orders at 4 K across its open band), which would demand ≈3-nm cells [6]. (D) Decision-noise budget versus temperature: vacuum, thermal bath, comparator amplifier noise, and threshold-band terms (Section 5).

4. The computing model: quantum-limited interference

4.1 Gaussian rules on the fabric

Pulses are comb-locked coherent states; in the measured quadrature x (vacuum variance ½) a pulse of N quanta has mean $\sqrt{(2N)}$. Three operations exhaust the analog layer, each with an exact Gaussian rule: propagation loss η at temperature T mixes in the bath, $V \to \eta V + (1-\eta)(\bar{n}+½)$; a junction combiner with amplitude weights $(\sqrt{w}, \sqrt{(1-w)})$ superposes means and averages variances; and regenerative gain G adds the matched-temperature phase-insensitive amplifier noise, $V \to GV + (G-1)(\bar{n}+½)$. The last rule is Part I's noise model made explicit: referred to a matched input floor it reproduces $F = 2 - 1/G$ (Part-I Eq. (7)) at every temperature, and reduces to the Caves bound [9] as $T \to 0$. (Provenance, community review July 2026: the comparator floor computed here at 300 K, 2.775 dB, and Part I's 2.77 dB floor, evaluated at 353 K, are one closed form at two temperatures — near G ≈ +9.7 dB the floor moves only ≈ 0.06 dB per dB of gain — so the numerical agreement between the two manuscripts is structurally

forced, not an independent cross-check. The promoted derivation note makes this falsifiable: exported at 353 K, F_dec must reproduce the parent floor to float precision, 2.768939660565078 dB; below 150 K the two repositories' floors diverge by construction — this repo's saturated τ_q gives 2.7925 dB at 77 K against 2.8012 dB from the parent's unsaturated lifetime — a designed separation, not a discrepancy.) A further promoted derivation (July 2026) fixes the scope exactly: the noise figure carries no quantum correction at any plasmon number — the matched floor's $\hbar$ cancels identically, at the 38-quantum knee exactly as at the rail — so the quantum/classical separation of this Part is a temperature phenomenon ($T^* \approx 43.7$ K), never an occupation-number one. A second promoted derivation (July 2026) draws the structural corollary: because these rules are exactly linear and the noise figure carries no 1/N correction, the chain is amplitude-scale-free in N and can generate no intrinsic plasmon-number threshold at all — asked its own coherence question, when single-plasmon coherence becomes classically indistinguishable, it answers $N^* \approx 0.02$–9 depending on temperature and criterion, one to four orders below the mesoscopic ladder of Section 3.1 and with different parameter scaling, so no tuning reconciles the two. Every occupation scale in this Part — the knee (a classical solver output), the rail (a coded constant, Section 3.1) and N_{op} (a default under a launch-budget cap whose denomination is undefined, Section 5) — is imported, never derived here; the warm band's absolute variances carry a 0.31–0.36 dB vacuum share, classical-equivalent in structure though not to the 0.1 dB level. The loss and gain rules are verified in the released package against exact Fock-space quantum evolution (truncated-basis Lindblad for loss, two-mode squeezing for gain) to better than 0.01 % at the package's check points, which are not the operating occupation, gain and amplitude; the combiner rule has no Fock-space check (Appendix QA). One deliberate modeling choice, conservative in principle only: decisions are modeled as midpoint-threshold homodyne discrimination — what the comparator cells are designed to implement — rather than the optimal joint quantum measurement, whose Helstrom bound [15] would improve the error exponent by at most a factor of two for these level sets. On this fabric that bound buys little warm and is out of reach cold: in-model, for the binary decision of a loaded digital cell (≈75 quanta, the gated-launch loaded fixed point of notes/2026-08-01-gated-requirement-round-trip.md; Section 3.1), an ideal homodyne threshold sits within 1 % of the Helstrom error at 300 K, while at 4 K the Helstrom error is 4×10^{-34} against a homodyne 2×10^{-18} — a bound this fabric, with no photon counting, cannot reach (notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md). For a binary decision between two pure coherent states, no receiver built from Gaussian operations and classical communication beats simple homodyne detection [17] (external; shown for the phase-shift-keyed pair, to which an on-off pair maps by a displacement). Whether either result carries over to QMAC-1's three-level ladder is open.

Weights are programmable the way everything on this fabric is programmable: a junction's split ratio is a gate-map choice, and a weight's sign (phase) is a programmed delay — the plasmon speed under a segment is density-tuned through Part-I Eq. (1), so a half-wave of extra path is a voltage, not a mask change. The interference core therefore implements $y = \Sigma\, w_i\, x_i$ with amplitude weights of either sign: a full four-quadrant multiply-accumulate by design. No released call exercises a signed weight: every call passes a split ratio $w \in [0, 1]$ to the combiner's $\sqrt{w}$, $\sqrt{(1-w)}$, on non-negative input amplitudes (demonstrated, source read). In-model, Part III [18] (Section 6.1) finds that a signed tile — an interference counter with a coherent offset arm, summing two signed ternary digits (trits; five levels on four comparators) — errs 5.4×10^{-4} at 4

K and 3.3×10^{-2} at 77 K, against 3.5×10^{-7} and 1.3×10^{-3} for an unsigned three-input counter under the same quiet-launch conventions (both tiles capped at the coded rail, which no run delivers to a deciding input; notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md), and so keeps each weight's sign digital, in the select stage that forms its products; Part III's own second check of the listing behind that result leaves five rework items open (Part III, Appendix TA).

4.2 Two computed design rules

The analog core must be passive. The regenerative cell's linear window ends at the ≈38-quantum knee, so any pre-amplifier caps the analog signal exactly where thermal noise is most damaging. Computed head-to-head (Appendix QA): the pre-amplified variant of the tensor unit (knee-capped at N_op = 19) delivers 0.50 symbol error at 300 K and 0.10 even at 4 K — against 0.14 and 1.3×10^{-6} for the passive core at N_op = 400. Phase-insensitive gain before quantization loses at every temperature on this fabric; gain belongs after the decision, inside the decoder cells, which are designed as threshold devices and want the rail anyway (Section 5). This is the quantum restatement of Part I's own discipline: restore digitally, never amplify analog. Note where the ≈38-quantum knee is load-bearing: here, and in the decoder sizing of Section 5 — in the classical machine design, that is, and not in the no-go of Section 3.3 (promoted community derivation, July 2026). The rule therefore inherits the open absolute calibration of Eq. (Q1) along with the knee itself; the head-to-head margin is orders of magnitude wide, so a factor-of-two shift in the knee would not move the verdict, but the dependence is real and is stated rather than assumed away.

The launch must be quiet. A launch at Part I's 20-dB classical SNR — modeled, as the released chain codes it, as the launcher's own amplitude jitter at 10 % of the full logic amplitude, additive and symbol-independent, on both arms — pins the 2-bit decode error between 3.4×10^{-3} (4 K) and 1.9×10^{-2} (77 K) at and below 77 K; warmer, the error reads 5.166×10^{-2} at 150 K, 0.1602 at 300 K and 0.2000 at 353 K, against 0.138 at 300 K with a quiet launch, and above ≈152 K the thermal term owns it (demonstrated; notes/2026-08-25-what-the-20db-launch-budget-denominates.md §1, §3). Classical launch noise does not cool away: no temperature takes the coded wall below 3.36×10^{-3}, its 4-K value to three figures (Figure Q3, grey curve). Part I's re-sync law consumes its 20 dB as a link budget against the thermal floor (Part I, Sections 6.2 and 7.4), a different physical quantity; which of the two Part I's figure denominates, at which plane and under which launcher physics, is open, and at 4 K the wall's magnitude spans 239× across the launcher-physics and reference-plane readings alone (in-model replica arithmetic; same note, §4 P3, §5 item 1). Quantum-limited operation requires a near-shot-noise-limited launch, which the photonic chain of Part I does not currently promise. In bench units the requirement is one scalar, proposed in the same note (§3; N_op = 400, this Part's additive coding at the N_op plane): σ_rel, the pulse-to-pulse standard deviation of the launched logic-1 amplitude over its mean, at the in-fabric (post-coupling) launch plane in the slot band, over ≥ 10^4 slots ($S = -20\log_{10}\sigma_{rel}$ for a coded SNR S). The 2-bit error reaches 10^{-3} at some temperature only for σ_rel ≤ 8.16 % (S ≥ 21.76 dB; in-model: the released threshold algorithm with the launch term added), and at 4 K it stays within 2× of its quiet-launch floor only for S ≥ 34.69 dB and within 1.1× only for S ≥ 43.53 dB (demonstrated, released calls). At the coded 20 dB and 4 K, more pulse energy lowers the error only 7.3× below the 3.4×10^{-3} wall: as N_op grows the error approaches a launch-limited floor of 4.6×10^{-4}, the same from 4 to 300 K to within 1 %, at the coded comparator trim offset c = 0.289 and decoder sharpness k_dec = 16 (Section 5), or 3.05×10^{-4}

from the launch term alone (in-model). The launch model is amplitude-only; launcher phase noise is open, and so is the transfer of a bench measurement, made before the coupling, to σ_rel, defined after it, together with the SNR convention (same note, §5). The quiet launch is Part II's sharpest new hardware requirement and is carried as an open item alongside Part I's launch-coupling efficiency.

5. QMAC-1: the minimal tensor unit, decoded to 2 bits

The smallest complete tensor operation is one weighted sum of two inputs, quantized. QMAC-1 (Figure Q2) is that unit on the fabric: two comb-locked input pulses (N_op quanta each at logic 1, in-fabric after launch coupling; N_op = 400 by default, capped by the launch budget) pass gate-programmed weight segments, interfere in one junction combiner, and the output amplitude is split to two comparator cells — threshold cells by design, with staggered, trimmed thresholds and sharpness k ≥ 16, which is a Part-II requirement whose achievability on a Part-I cell is open (Table QB1) — whose thermometer code is mapped to binary by one inverting cell and one AND cell (both qualified below). What the launch-budget cap on N_op denominates is not defined: read as Part I's 20-dB budget, 400 quanta lies 18.41 dB over $\bar{n}$ at 300 K in the quanta convention ($N/\bar{n}$), below the budget, and 21.06 dB in the quadrature convention ($2N/(\bar{n} + ½)$, the squared quadrature mean over its variance), above it (demonstrated, arithmetic; notes/2026-08-25-what-the-20db-launch-budget-denominates.md §1). Five active cells, ≈12×6 lattice sites (≈13×14 μm at the Part-I site pitch, Figure Q5), one MAC per 4-ps slot as the design baseline, wave-pipelined like everything on this fabric — with Part I's timing qualification inherited: once per-slot flushing is required against cavity ring-down, fan-out-2 logic closes in-model near 0.1 THz rather than at the 4-ps slot (Part I, Section 6.2), a figure that survives only under the pulse-knee convention (the knee of a released 3-cycle pulse), which the released solver rules out for slot-filling drives, because the compression knee migrates with drive duration (notes/2026-07-31-physical-launch-gated-frontier.md). With a physical launch confined to its slot and gated by the data (logic-0 slots left undriven), the fan-out-2 frontier is 69 GHz (band 63–74 GHz), a 14.5-ps slot (in-model, adjudicated with the cell biased at 0.7 of the analytic threshold at 353 K; same note). At the ratio bias proposed for the bench (0.7 × the measured threshold) it is neither confirmed nor moved: the residual floor in logic-0 slots becomes the binding constraint there, and its specification must be imported (open; notes/2026-08-01-ratio-bias-gain-table.md, notes/2026-08-02-zero-floor-spec-dissolved.md; Part III [18], Section 2.2). The slot-rate throughput figures below are therefore baseline figures.

By design, the decoder is not new hardware. Comparators, inverter, AND: these are the Part-I gate vocabulary, and the block is cell-for-cell the Part-I half adder with its input stage re-trimmed as a two-level flash ladder. In the digital special case (both weights ½, inputs ∈ {0, N_op}) the interference levels are 0, N_op/2 and 2N_op — equally spaced in amplitude — and the decoded word b_1b_0 is the binary count A+B: sum and carry (Table Q2). The Part-I half adder re-emerges as the decoder of a single quantum interference — the demonstrator of Part I and the decoder of Part II are the same five cells, re-programmed. That framing carries two qualifications. The first falls on the comparators: k is defined in Part I against the cell's rail-to-rail input swing, which is ≈5× the ladder range this decoder works on, so k_dec ≥ 16 is not reachable in-model by re-trimming a Part-I logic cell, and the two ladder comparators are a redesign rather than a re-trim. The second falls on the inverter and the AND, which carry over

only as Part I states them (Part I, Section 5): in-model the released cell supplies neither as a threshold gate (notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md). For a carrier-frequency input, as the decoder's are, the released cell's only density coupling, rectification, accumulates rather than depletes, with a sharpness of only 0.18–0.26 on Part-I Eq. (5)'s tanh scale at the delivered swing; two in-phase logic-1 inputs exceed one by only 2.6–3.3 dB (demonstrated, in the retuned configuration of Section 3.1, in runs whose fluid speed exceeds graphene's v_F, which the released solver does not cap), so the cell compresses rather than thresholds; and in the amplitude encoding this decoder uses, the only inverter the cell itself admits is destructive interference with a coherent clock pulse, of sharpness 2.9 on that tanh scale, which reaches 10 dB of extinction only within 10.1–12.3° of phase error (demonstrated, by grid, tightening under refinement) and is at best marginally restoring — under phase encoding a NOT is a half-period delay (notes/2026-10-17-a-pumped-cell-restores-phase-in-a-narrow-window.md). The released decoder treats its NOT and AND as ideal digital cells (demonstrated, source read: qdecode.py), so no released number moves; how the decoder's inverter and AND are to be built is open, like its comparators.

A	**B**	**interference level (quanta)**	**t_2 t_1**	**b_1 b_0**	**= A+B**
0	0	0	0 0	0 0	0
0	1	N_op/2	0 1	0 1	1
1	0	N_op/2	0 1	0 1	1
1	1	2 N_op	1 1	1 0	2

Table Q2. QMAC-1 digital truth table: the 2-bit decode of one interference is the half adder of Part I (b_1 = carry, b_0 = sum). Verified digitally in the released package.

In analog mode the same unit computes $y = \sqrt{w}\cdot x_A + \sqrt{(1-w)}\cdot x_B$ and the ladder quantizes y to 2 bits. The per-shot analog precision — effective bits of one MAC against the full noise budget — is 0.6 bits at 300 K, 2.0 at T_Q and 2.5 at 4 K at the default N_op = 400 (Figure Q4A). The measure grows with pulse energy and is bounded above by the decoder's own conventions, not by quantum noise: it stays below $\log_2(k_dec/(2\sqrt{3}\cdot c))$ (in-model, exact algebra of the released qmac.analog_enob), exactly 4.000 bits at the released k_dec = 16 (a margin design rule) and $c = 1/\sqrt{12}$ (an open hardware parameter). At N_op = N_rail/2 ≈ 1916 the same function returns 1.7, 2.7 and 3.3 bits at 300, 77 and 4 K (demonstrated, released qmac.analog_enob at N_op = 1916), but that is an amplitude the coded rail allows, not one the launch budget or any delivered swing reaches (notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md). At N_op = 400 this fabric is a 1–2-bit-per-shot analog machine, which is why the decoder is sized at 2 bits and why lower-bit decoding is not a concession but the design point. A worked weighted example (w = 0.7, 5×5 input grid, 77 K) holds worst-case rms noise at 0.36 LSB of the 2-bit ladder (Appendix QA). Deeper precision can be bought with slots, within a hard topology limit (notes/2026-07-16-no-physical-avg16-accumulator.md): accumulating 16 slots in the electronic charge-memory layer (Part I, Section 6.4) and taking one decision on the accumulated mean averages the per-slot propagation noise as 1/16 — mean accumulation, not a majority vote over per-slot decisions, which scales far more weakly (a 9-of-16 vote is ≈175× worse at 300 K). The only scheme that would also divide the comparator's own amplifier noise is a lossless coherent combiner of the 16 slots ahead of one decision, and that is unbuildable here — it needs the earliest slot held for 60 ps against a ~1 ps lifetime (Part I, Section 6.4: no native plasmonic storage). The buildable

accumulate-then-decide machine charges the comparator amplifier noise once, undivided, so averaging is a cold-class instrument — a ≈2.5× improvement at 300 K, reaching a 10^{-6} error only below ≈45 K, at 1/16th throughput (still 1.6×10^{10} MAC/s per unit at the slot-rate baseline; Section 5's timing qualification applies) — not a room-temperature buy. Two decoder facts complete the picture. The static threshold term is not a decision variance: the tanh transition band of Part-I Eq. (5) is a deterministic, strictly monotone transfer, so it carries the decision event over unchanged ($\{A_{out} > \theta\} = \{A_{in} > f^{-1}(\theta)\}$), moves no threshold, and adds no noise; what finite k sets is the incremental gain at threshold, a latency and margin parameter. The static term that does survive is the comparator's own input-referred offset after trim, $V_{static} = c^2(x_{range}/k_{dec})^2$ with c the rms offset as a fraction of the transition width — a property of the trim loop and downstream mismatch, not of k; the released chain fixes $c = 1/\sqrt{12} = 0.289$, a value nothing in either manuscript derives or bounds, so c is an open hardware parameter (a QG2-family bench measurement) and the published q2bit_avg16 column is the c = 0.289 sensitivity case, unchanged to every digit. And the $k_{dec} \geq 16$ rule stands on geometry rather than statistics: the tangent-extrapolated band $2\cdot x_{range}/k$ must clear the ladder's level gap $x_{range}/2$, so $k_{dec} > 4$ is necessary and k = 16 preserves 75 % of the half-gap where k = 8 preserves 50 % — a margin choice, not a derived optimum, and its achievability is a separate open question (Table QB1).

Figure Q2. The minimal quantum-analog tensor unit and its 2-bit decode (N_{op} = 400 quanta)

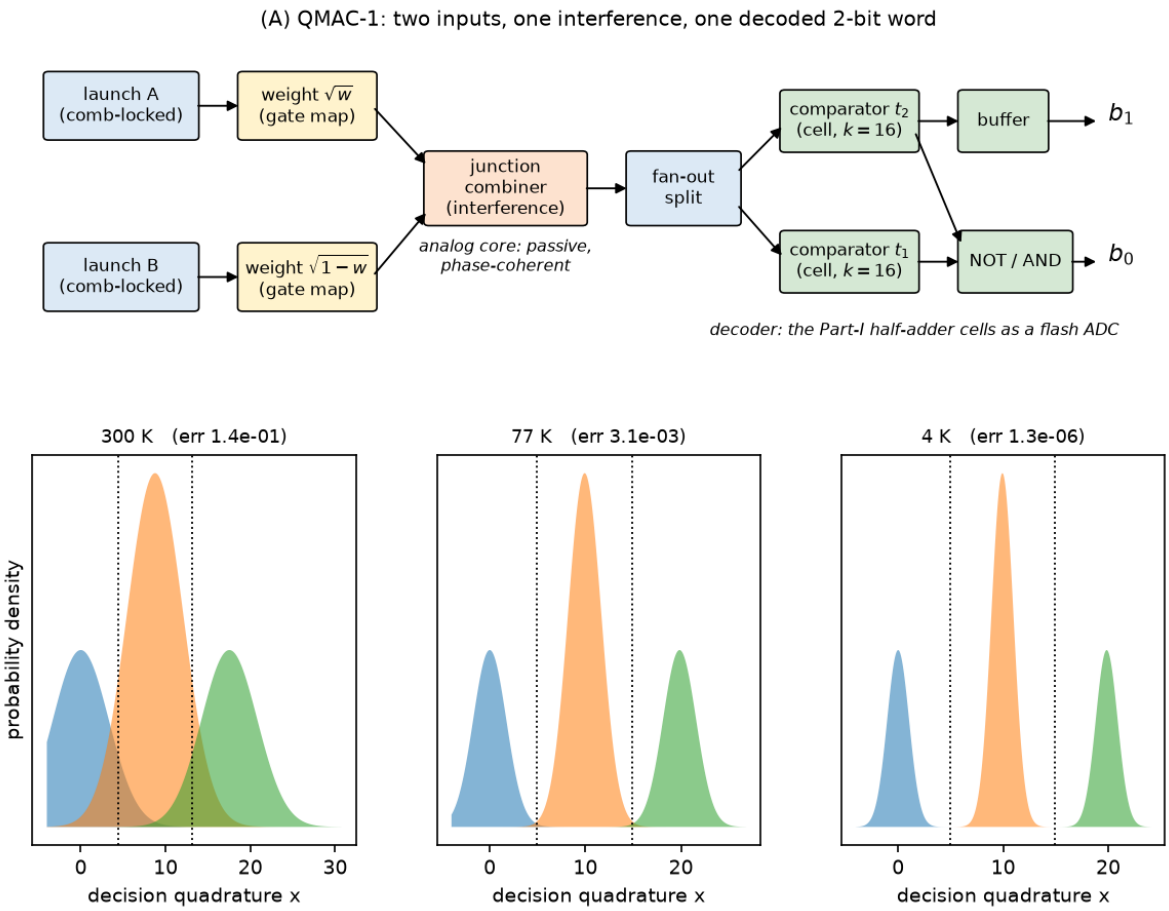


Figure Q2. QMAC-1. (A) Cell-level netlist: comb-locked launches, gate-programmed weights, one junction combiner (the interference that computes), fan-out split, two comparator cells, and the thermometer-to-binary cells; the decoder is the Part-I half-adder block re-programmed (comparators redesigned; inverter and AND as Part I states them; Section 5). (B) The three interference levels and their Gaussian noise at 300, 77 and 4 K with midpoint thresholds (N_op = 400): the same levels drown at 300 K, separate at 77 K, and sit an order of magnitude apart at the vacuum floor.

Figure Q5. QMAC-1 on the gate lattice: ~12 x 6 sites (~13 x 14 um at the Part-I site pitch), five active cells - the Part-I block, re-programmed

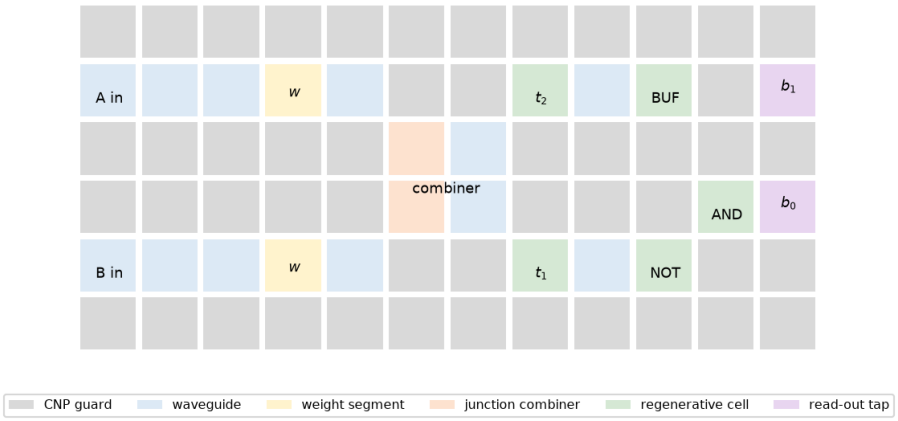


Figure Q5. QMAC-1 on the gate lattice (Part-I Figure 5b convention): ≈12×6 sites including guards, five active cells, weights and combiner as programmed junction segments, read-out taps at the boundary. By design, the map is a re-programming of the Part-I half-adder block, not a new tile (comparators redesigned; inverter and AND as Part I states them; Section 5).

6. Error rates versus temperature

Figure Q3 and Table Q3 put the quantum-analog unit and the classical digital fabric on one temperature axis, 4–400 K, holding the hardware fixed and letting only $\bar{n}(T)$, $\tau(T)$ and the loss/gain budget move. The classical-digital curve is this package's own construction (Part I never quotes a bit-error rate): a rail-restored decision — swing at the coded ≈10 % rail (≈3833 quanta; anchored at neither reference plane — Section 3.1), Part I's static noise margin (0.256 of swing), against the matched-temperature amplifier noise of one restoring cell. It is the physics floor of a digital fabric whose cells meet that margin at the coded rail swing, with technical noise excluded on both sides of the comparison. Both premises are qualified. The swing: no run of the record delivers it to a deciding input, and the released decision law needs ≈2924 input-referred quanta, 0.76 of the rail, to reach 6.3×10^{-9} per decision at 300 K, the pass line of Part III's gate DG1 [18] (in-model, with the decision noise taken, as released, at the matched-temperature amplifier rule; notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md). The margin: 0.256 is the released closed-form rule cell.static_noise_margin at Part I's k = 8, 10-dB-extinction design point (Part I, Section 5); the rule never evaluates the cell's transfer, and this column's margin constant is its four-decimal copy, so the classical column inherits Part I's margin rather than independently witnessing it (demonstrated, source read and arithmetic; notes/2026-08-02-zero-floor-spec-dissolved.md). In-model the released cell does not deliver that design rule (Section 5; notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md): its interferometric inverter reaches only 2.9 on Part-I Eq. (5)'s tanh scale and is at best marginally restoring.

T (K)	n̄	classical digital BER	quantum-analog 2-bit	quantum-analog 1-bit	2-bit, 16-slot avg	analog bits/shot
353	6.87	9.1×10^{-10}	1.8×10^{-1}	9.6×10^{-3}	6.5×10^{-6}	0.43
300	5.76	3.6×10^{-11}	1.4×10^{-1}	3.9×10^{-3}	4.7×10^{-7}	0.62
150	2.65	2.4×10^{-20}	2.5×10^{-2}	1.1×10^{-5}	2.2×10^{-13}	1.29
77	1.16	5.7×10^{-37}	3.1×10^{-3}	4.8×10^{-9}	2.6×10^{-19}	1.73
48	0.58	1.8×10^{-55}	3.9×10^{-4}	1.9×10^{-12}	1.6×10^{-23}	2.00
20	0.10	2.3×10^{-98}	6.5×10^{-6}	2.9×10^{-19}	5.3×10^{-29}	2.36
4	6.2×10^{-6}	$<10^{-99}$	1.3×10^{-6}	6.3×10^{-22}	1.7×10^{-30}	2.46

Table Q3. Error per operation versus temperature (N_op = 400 quanta per input; classical digital at the coded rail swing, which no run of the record delivers to a deciding input — Section 6). Every column is recomputed by run_all.py in the released package, with a quiet launch (Section 4.2) and, in the quantum-analog columns, the decoder amplifier's idler at the lattice occupation. On the idler axis alone the table is optimistic or exact under every classical mechanism notes/2026-08-05-idler-added-noise-and-the-nf4db-requirement.md names: at 300 K the 2-bit entry, 1.4×10^{-1} (0.138 to three figures), would be 0.205 with an idler at 2n̄, and 5.0×10^{-2} only at the Caves floor of a vacuum idler (demonstrated; that note, §4); a separate correction — how the released chain pairs the decoder amplifier's noise with its gain — is undetermined in both sign and magnitude (that note, §6).

Reading the table downward: at the Part-I band the classical digital fabric is essentially error-free in this idealization (3.6×10^{-11} at 300 K, at the coded rail swing and Part I's k = 8 design margin, both qualified above) while the 2-bit quantum-analog decode is unusable per shot (0.14) — digital restoration wins at room temperature by restoring at ≈10× the analog unit's per-input quanta and giving up linearity. That win holds as stated at the coded rail swing only. At the loaded fan-out-2 fixed point composed in Section 3.1 on the absolute and ratio legs (99–277 quanta), the same decision law errs 0.040–0.147 per decision at 300 K, within ×3.5 of the 2-bit decode's 0.14 on either side, and 0.18 at the gated-launch loaded fixed point of notes/2026-08-01-gated-requirement-round-trip.md, where the comparison reverses (both swings computed at 353 K and paired here with 300 K noise); a per-operation comparison, several decisions per half adder, would need its own accounting (in-model, decision noise as released; notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md), and the k = 8 margin makes these digital figures optimistic for the released cell (above). The quantized noise floor turns rail-scale restoration from a choice into a requirement: had the digital fabric restored its levels at the 1-dB knee (≈38 quanta) instead of the rail, its decision would be thermal-noise-limited to ≈0.26 BER at 300 K — a constraint on Part I that only becomes visible once the mode is quantized, and one its 20-dB launch-SNR budget, read as a link budget against the thermal floor (≈600 quanta over it; Section 4.2), implies only weakly: at the budget level itself — 576 quanta at 300 K counted as 100 n̄ (the quanta convention of Section 5) and 626 counted as 100 (n̄ + ½) — the released decision law still errs 5.8×10^{-3} and 4.2×10^{-3} per decision respectively. The requirement holds as a property of the released law (0.257 at the published knee, reproduced); the fabric that meets it does not appear in the record (in-model; same note). The 1-bit decode, spending the full decode range on one boundary, is already serviceable warm (3.9×10^{-3} at 300 K, 9.6×10^{-3} at the 80 °C cap; conditional, like every warm gain figure, on the warm DS threshold of Section 1) — the quantitative content of "lower-bit decoding is acceptable," and the warm story this Part actually stands on. The 16-slot averaging column is a cold-class instrument, not a warm one, and the distinction is a topology fact. The averaging path holds no vote and no per-shot decision

(notes/2026-07-14-q2bit-avg16-averaging-convention.md): 16 slots accumulate as an analog mean and one Gaussian midpoint decision is taken on it. The released column q2bit_avg16 (4.7×10^{-7} at 300 K) is the noise algebra of that decision with every per-slot term divided, V_{16} = (V_signal + V_decoder-amp)/16 + V_static — which is exactly the lossless coherent combiner of Section 5, unbuildable on this fabric (notes/2026-07-16-no-physical-avg16-accumulator.md). The machine that can be built accumulates as charge and charges the comparator's amplifier noise once, undivided, giving $\approx5.6\times10^{-2}$ at 300 K and reaching 10^{-6} only below ≈45 K: averaging converts the 2-bit channel from "never 10^{-6} at any temperature" to "10^{-6} in the cold class," which is real and useful but is not a room-temperature buy. Within the released bookkeeping the static term V_static = c^2(x_range/k_dec)^2 is the comparator's input-referred trim offset, not the tanh band (Section 5), with c open and coded at $1/\sqrt{12} = 0.289$; its $n \to \infty$ limit is the closed form p_floor(c, k) = (¾)·erfc(k/(4√2·c)) = 8.7×10^{-44} at c = 0.289 and the k = 16 rule, scoped to the equally spaced digital levels of Table Q2 (weighted operation breaks the cancellation). Both that column and its floor are contingent on k and c jointly: at k = 8 the 300 K figure degrades 37× and the floor rises to 3.2×10^{-12}. Two limitations stay open regardless of topology — whether the charge-memory layer delivers 16 independent slot samples (comb-clock, gate 1/f and thermal drift inside the 64-ps window would partition into the non-averaging term — and cavity ring-down carries pattern memory between slots — the binding coupling of this kind in-model) and whether the trim residual is truly slot-static. Per-shot 2-bit operation is a cold-class mode: error 10^{-3} needs ≤59 K, and the 1-bit decode reaches 10^{-6} by 117 K and 10^{-9} by 69 K, all with a quiet launch and the decoder amplifier's idler at the lattice occupation — two load-bearing conditions. The cut is a statement about the idler, not the lattice: with a vacuum idler it is 105.7 K, at one idler quantum 45.8 K, and from 1.5 idler quanta 10^{-3} is reached at no temperature the released code can evaluate, down to 0.0677 K (demonstrated; notes/2026-08-05-idler-added-noise-and-the-nf4db-requirement.md §4). The launch adds an independent cut, through a temperature-invariant noise floor: launcher amplitude jitter lowers the 10^{-3} temperature (58.1 K at S = 40 dB, 52.7 K at 30 dB, 33.8 K at 24 dB), and below S = 21.76 dB (σ_rel above 8.16 %) no temperature reaches 10^{-3} (in-model: the released threshold algorithm with the launch term added; N_op = 400, this Part's additive coding; notes/2026-08-25-what-the-20db-launch-budget-denominates.md §3). All of these assume the optimistic end (−1 dB) of Part I's −1 to −3 dB junction band; at −3 dB junctions the 77 K 2-bit error degrades to 5.6×10^{-2} and the 77 K 1-bit error to 1.8×10^{-4} — junction quality is a first-order lever on every cold-class number, and the sensitivity is reported alongside the defaults in the released results.

The temperature structure has two landmarks, and they must not be conflated. T_Q ≈ 48 K is where the noise density crosses from thermal- to vacuum-dominated ($\bar{n} = 0.58$ there — still comparable to the vacuum ½). The error curves, being erfc tails, keep falling steeply until $\bar{n} \ll ½$ and saturate at their vacuum-set floors below ≈15–20 K (decoder idler at the lattice occupation, quiet launch): 1.3×10^{-6} (2-bit) and 6.3×10^{-22} (1-bit) at N_op = 400, the 2-bit floor falling to 1.3×10^{-10} at N_op = 800 (Figure Q4B). The floor's value is an engineering choice of pulse energy, but its temperature-independence is not: a noise floor that flattens through the T_Q knee and stops improving below ≈15 K is the standard quantum limit made visible on a graphene chip, and Section 8 makes its observation a pre-registered gate. The grey curve is the cautionary tale: with the coded 20-dB-SNR classical launch (Section 4.2), no temperature takes the 2-bit error below 3.36×10^{-3}, its 4-K value to three figures.

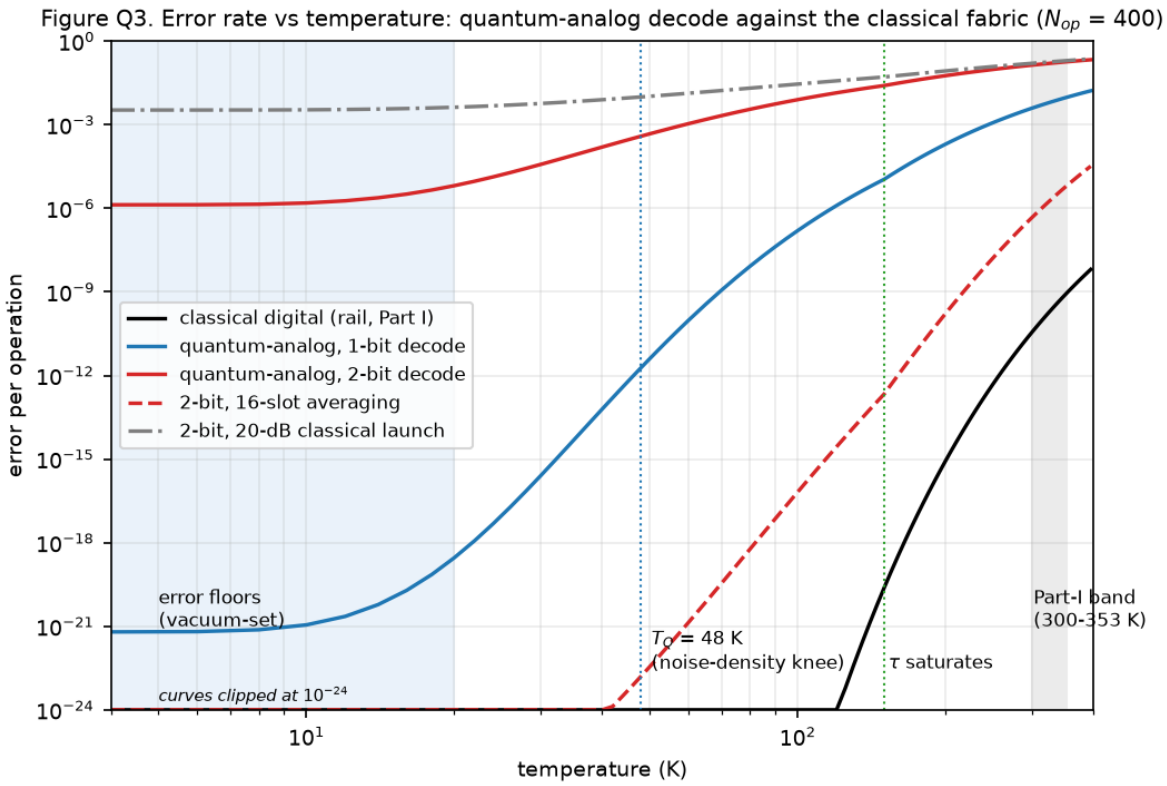


Figure Q3. Error per operation versus temperature (N_op = 400; quiet launch except the grey curve; decoder idler at the lattice occupation): classical digital fabric at the coded rail swing (black; Section 6), quantum-analog 1-bit (blue) and 2-bit (red) decodes, 2-bit with 16-slot averaging (dashed), and the coded 20-dB classical-launch wall (grey; Section 4.2). Markers: T_Q = 48 K (the noise-density knee), 150 K (lifetime saturation), the Part-I 300–353 K band (shaded), and the sub-20 K region where the quantum-analog curves sit on their vacuum-set floors — the quantum limit as an observable.

Figure Q4. The analog precision budget

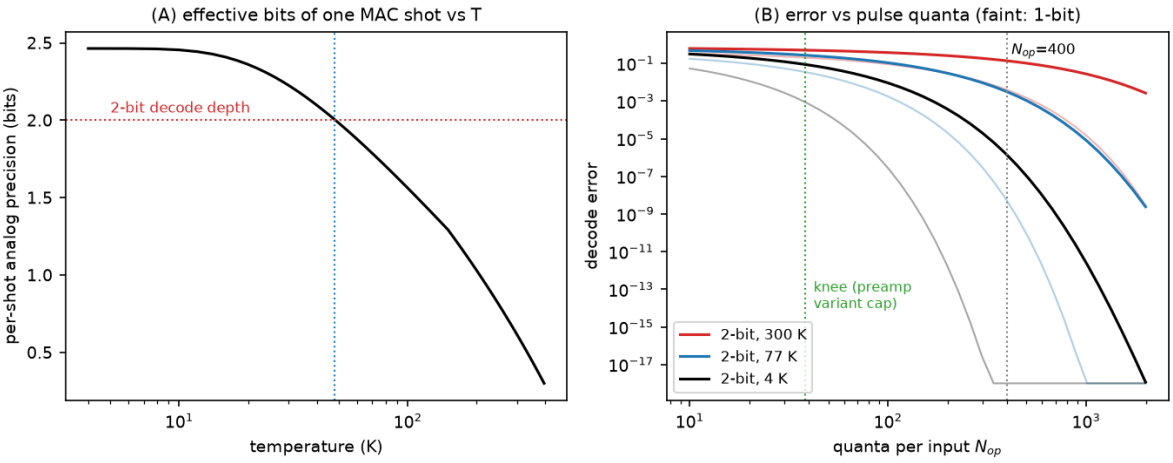


Figure Q4. The precision budget (quiet launch; decoder idler at the lattice occupation). (A) Per-shot analog precision of one MAC versus temperature at N_op = 400: ≈0.6 bits at 300 K, 2 bits at T_Q, 2.5 bits at the floor — the design point that sizes the decoder at 1–2 bits, below a 4.000-bit ceiling set by the decoder's own conventions (Section 5). (B) Decode error versus quanta per input at 300/77/4 K (faint: 1-bit): the knee (38 quanta) caps any

pre-amplified variant; the passive core scales with pulse energy up to the launch budget, a cap whose denomination is undefined (Section 5), with the 2-bit floor reaching 1.3×10^{-10} at N_op = 800.

7. What is, and is not, quantum here

Is: the noise. Below T_Q the decision noise becomes vacuum-dominated — the machine computes against $\hbar\omega/2$ per mode rather than a thermal bath — and, with a quiet launch and the decoder amplifier's idler at the lattice occupation (Section 6), from ≈15–20 K down its error floors are set by quantum mechanics (vacuum variance, the Caves cost of the decoder's gain), not by temperature. The per-shot information of one pulse is shot-noise-bounded at a given pulse energy (Figure Q4A). These are quantitative, falsifiable quantum statements, and gate QG2 tests the sharpest of them.

Is not: the computation. Coherent states, beamsplitters, phase-insensitive amplifiers and homodyne decisions are all Gaussian; a Gaussian machine with coherent-state inputs is efficiently classically simulable, and universal continuous-variable quantum computation requires exactly the non-Gaussian resource [7, 8] that Section 3.3 shows this cell cannot supply ($\chi/\kappa \approx 10^{-5}$). No entanglement is generated or claimed anywhere in QMAC-1. Nor is the mesoscopic ladder itself a quantum result: 38 and 3833 are a classical Part-I solver amplitude and a coded rail constant (Section 3.1) converted into quanta by Eq. (Q2), and the Gaussian noise chain — being exactly linear and amplitude-scale-free in N — contains no plasmon-number threshold of its own from which they could be derived (promoted community derivation, July 2026). What is quantum here is the noise the ladder is measured against, not the ladder. The honest name for this machine is quantum-limited analog computing: classical arithmetic executed, at a given pulse energy, at the precision ceiling quantum mechanics assigns to unentangled light.

The upgrade paths are named, with their prices. (i) Blockade cells: ≈3-nm structures at $\chi/\kappa \approx 1$ [6] — a different lithography universe, and lifetime-limited even then. (ii) Squeezed operation: a DS cell driven parametrically at $2f_0$ is a phase-sensitive amplifier whether the modulated bias is read as the gate, at the source density clamp, or as the drift, at the drain current clamp (in-model, on the released solver's cell in the retuned configuration of Section 3.1 at 353 K; notes/2026-10-17-a-pumped-cell-restores-phase-in-a-narrow-window.md). A $2f_0$ tone at 5 % of the source density gives 8.9 dB (N = 720 grid) to 9.7 dB (N = 240) of phase-sensitive contrast on the drain density on the ratio bias leg, against 3.1–3.2 dB for the same cell without its DC bias; a 50 % modulation of the drain current gives 16.5 dB (N = 240 grid, a large-signal reading); a pump on the body force that sets the drift couples only weakly (0.51 dB on the drain density at a 0.2 modulation, in one assessor seat's rows that the note records, not in its listings, against 6.24 dB for a 0.2 modulation of the drain current in the note's own Table P; in-model), so the drift reading's coupling lives in the drain clamp. The admissible window is narrow: the amplifier restores phase at knee-level swings, where the fluid speed stays below graphene's v_F, and only weakly at the swing a loaded cascade delivers; above an onset that rises with grid refinement the drain route latches as a parametric oscillator which, gated in the slot of Section 5, amplifies a bit stream's flips but re-levels none. Under Part I's Eq. (1) the source route would need ≈2.3× the density modulation for the same coupling (in-model estimate), and whether a contact can impose, at 2 THz, the 70–85 % drain-current modulation at which the drain route latches is open. An ideal phase-sensitive amplifier evades the $F = 2 - 1/G$ floor [9] and could push

decisions below this Part's standard quantum limit, the vacuum-set floor of Section 6, which includes the Caves cost of the decoder's gain — though for a binary decision between two pure coherent states not below the homodyne limit of Section 4.1 [17]; for this cell that noise consequence is named, not computed, and squeezing is not run. Graphene plasmon squeezing has no experimental precedent and is listed as an open item, not an ingredient. (iii) Entanglement distribution across the fabric: plasmonic channels preserve entanglement at optical frequencies [4]; nothing comparable exists at THz. None of these is assumed by any number in this Part.

8. Falsifiability: the bench gates

In Part I's discipline, every claim above reduces to pre-registered pass/fail gates on the same class of device the Part-I protocol already specifies, plus the quantum-specific read-out chain. The five gates are listed in Table Q4; QG2 is the existence proof of the quantum regime and the single most informative outcome, pass or fail.

Gate	Observable	Pass threshold	Model target
QG1	two-pulse interference visibility at a junction combiner (comb-locked launches)	≥ 90 %	≥ 99 % (phase budget 42 fs vs 1.4 fs, the ≈9-mrad RMS residual reported for an optical phase lock, expressed at 1 THz — external, an optical-frequency system, not a THz clock; Part I, Section 3.2)
QG2	read-out noise floor vs temperature, 300 → 4 K	departs from the classical ∝ T line by ≥ 40 % at 20 K and flattens below ≈15 K	floor $\bar{n} + ½ \rightarrow ½$: knee at $T_Q \approx 48$ K, within 10 % of vacuum below ≈16 K
QG3	QMAC-1 2-bit symbol error at 77 K, $N_{op} \approx 400$	$\leq 10^{-2}$	3.1×10^{-3} at −1 dB and 5.6×10^{-2} at −3 dB junctions, both with a quiet launch and the decoder amplifier's idler at the lattice occupation; 1.9×10^{-2} at −1 dB under the coded 20-dB launch (Section 4.2)
QG4	1-bit (full-range) decode error at 300 K	$\leq 10^{-2}$	3.9×10^{-3}, with a quiet launch and the decoder amplifier's idler at the lattice occupation, and conditional on the warm DS threshold (Section 1)
QG5 (stretch)	cross-Kerr phase from a 400-quantum control pulse, averaged	bound $\chi/2\pi$ within the 0.13–2.6 MHz band	6.6 mrad per 2 ps, ≈15 shots to resolve

Table Q4. Pre-registered gates for the quantum extension. QG1 and QG2 precede the tensor unit: without visibility there is no analog computer, and without the noise-floor saturation there is no quantum regime to claim. QG3 is deliberately a joint test of the decode chain and the −1 dB junction budget — at −3 dB junctions the model itself fails the gate, so the Part-I junction-variant die gates QG3. QG3 is likewise conditional on the launch: its model target assumes a quiet launch, and under the released chain's coded 20-dB launch (σ_{rel} = 10 %, additive) the model itself fails the gate, at 1.9×10^{-2} for −1 dB junctions (demonstrated; notes/2026-08-25-what-the-20db-

launch-budget-denominates.md) — σ_rel, the bench scalar of Section 4.2, names that condition. QG4's model target rests on the same quiet launch, on the decoder amplifier's idler at the lattice occupation (which QG3's also assumes; Table Q3 caption), and on the warm DS threshold that Section 1 leaves open. All gates assume the Part-I calibration sequence (σ, τ(T), s, M_th, coupling) has run first.

Two dependencies are inherited from Part I and stand in front of everything here: the launch-coupling efficiency into the acoustic mode (unquantified there — photoconductive launch into gated graphene plasmons has been shown experimentally only at 4 K [16], an external fact — and Part II additionally requires the launch to be near shot-noise-limited: in bench units $\sigma_{rel} \leq 8.16$ % merely for the 2-bit error to reach 10^{-3} at some temperature, and a launch SNR $S \geq 34.69$ dB for the 4-K error to stay within 2× of its quiet-launch floor; Section 4.2), and the Boltzmann–Maxwell tier, which now also owns the Kerr coefficient c_K — and with it the width of the "five orders" gap, ≈4.5–5.8 orders at 4 K across the $c_K = 0.05$–1 band — the absolute calibration of Eq. (Q1) that fixes the knee and the rail in quanta, the absolute noise calibration, and every cold-side gain number below the ≈255 K line above which the kinetic closure has been evaluated; and, above that line, whether the warm DS threshold itself exists under finite-temperature kinetics, which that closure does not settle (in-model none at 353 K for an electron–electron rate prefactor $A \leq 1$ in the single-rate operator; open for graphene's own electron fluid; Section 1).

9. Conclusion

Part I asked whether room-temperature graphene can gain back what a plasmon loses; this Part asked what the same fabric is worth as a computer when its pulses are counted in quanta. The answers are symmetric. Downward, to the single quantum: nothing — the Kerr gap is five orders of magnitude at the carried Kerr coefficient (≈4.5–5.8 orders at 4 K across its open band), the geometry independently forbids a 1-THz blockade cell, and the fabric will host no qubit. Sideways, at tens to thousands of quanta: a genuine machine — a phase-coherent interference core that multiplies and adds in one junction, decoded to the 1–2 bits per shot that shot noise affords at the design pulse energy, by the five-cell block of Part I's half adder as designed, with the two ladder comparators redesigned for the decoder's $k \geq 16$ and the inverter and AND carried over from Part I, though in-model the released cell supplies neither as a threshold gate (Section 5). Applied to the adder tree of a ternary language model, the core is not a route to a better dot-product lane, Part III's fixed datapath of cells that forms one 32-wide block sum of a dot product per slot (in-model; Part III [18], Sections 6–7, five rework items of its listing open): its interference counters would save at most about a third of the lane's cells, and only cold, where, under Part III's booking A0 (the released decision law of Section 6) and with cold-cell dissipation uncertified, a 77 K lane needs a cooler at about 40 % of Carnot or better to break even (39.6 % for Part III's hybrid lane, 41.4 % all-digital), and 20 K and 4 K lanes lose even at Carnot; Part III keeps the digital lane, and the three-input counter as a test tile beside the QG gates. The temperature budget is the honest price list, read with a quiet launch and the decoder amplifier's idler at the lattice occupation (Section 6): 1-bit decoding at the Part-I band with no cryogenics (conditional, like every warm gain figure, on a warm DS threshold whose existence is open: in-model the single-rate kinetic operator finds none at 353 K for an electron–electron rate prefactor $A \leq 1$ and one at 300 K only at $A = 3$ with $\tau_E \to 0$, Section 1; 16-slot averaging is a cold-class instrument, reaching 10^{-6} only below ≈45 K), per-shot 2-bit operation from ≈60 K down, and below ≈20 K a floor that cooling cannot improve — which is not a defect but the standard quantum limit, and the most falsifiable statement this Part makes. The model chain is

released, the gates are pre-registered, and the design is now exactly as falsifiable as Part I was built to be.

Appendix QA. The quantum model chain: conventions and released code

The released package fable-model-quantum mirrors Part I's fable-model-chain: pure Python, NumPy only (Matplotlib for figures), every module independently runnable and self-checking, one driver (run_all.py) reproducing the package's results and writing them to results.json, and figures.py regenerating Figures Q1–Q5. Conventions: x-quadrature with vacuum variance ½ and coherent mean $\sqrt{(2N)}$; thresholds at trimmed midpoints; all losses at the Part-I budgets (−1 dB junctions, −3 dB splits, λ_p/2-convention propagation); decoder comparators at k = 16 with F = 2 − 1/G noise at the working bias. Two conventions were made explicit by promoted July 2026 notes and are recorded here. (i) Provenance of the mesoscopic ladder (notes/2026-07-13-knee-rail-not-derivable-from-noise-chain.md, promoted 3-of-3): the compression knee ε = 1 % and the rail ε = 10 % are hardcoded classical Part-I constants living in qmode.py alone (the knee a solver output; the rail a coded value the record marks assumed, never produced by any run — notes/2026-10-08-no-rail-scale-one-at-deciding-inputs.md) — inputs to the quantum chain, not derived anywhere in its noise chain — and no module of that chain produces a plasmon-number threshold. $\varepsilon_1 = 1.6153\times10^{-3}$, N_knee = 38.326 and N_rail = 3832.6 reproduce from Eqs. (Q1–Q2) against an independent hand reconstruction to ≈1.2 % in ε_1 and, since $N \propto \varepsilon_1^{-2}$, to ≈2.4 % in the two quanta counts; the residual is traced to the plasmon speed, since qconstants.py computes s from the RPA form, 2.331×10^6 m/s, ≈1.2 % above the quarter-wave $4Lf_0$ a reader would infer from Section 3.1. Both s and $v_F = 10^6$ m/s are pinned in qconstants.py and stated here. (ii) Averaging convention (notes/2026-07-14-q2bit-avg16-averaging-convention.md, promoted 3-of-3): q2bit_avg16 is qmac.decision_variance's slot-mean accumulation with a single midpoint decision, V_{16} = (V_signal + V_decoder-amp)/16 + V_static, the static term evaluated by qdecode.threshold_band_variance at the post-loss decoder-plane swing (x_range ≈ 17.52 at 300 K, 19.85 at 4 K) — reproducing the published column to all digits. By the promoted July 2026 derivation, that module's coefficient is an open hardware parameter rather than a derived one: what it codes as $(x_{range}/k)^2/12$ is $c^2(x_{range}/k_{dec})^2$ at $c = 1/\sqrt{12} = 0.289$, so the column it feeds is the sensitivity at that value and not a consequence of the comparator's tanh band (Section 5). The convention string, the coefficient c, and the averaging floor in its generalised form $(¾)\mathrm{erfc}(k/(4\sqrt{2}\cdot c))$ — of which the published $(¾)\mathrm{erfc}(\sqrt{(3/8)}\cdot k) = 8.73\times10^{-44}$ at k = 16 is the $c = 1/\sqrt{12}$ case — are tracked as a WP1 deliverable for machine-readable results.json keys — the floor key jointly parameterized in c and k_dec, since a k-only key would re-assert a withdrawn dependence; c itself is a bench quantity of the QG2 family, not a chain output. The working-bias gain re-derives Part I's continuous-wave cavity transfer verbatim (self-checked against fable-model-chain at 300/353 K), so the decoder's noise figure is the single comparator's Eq.-(7) value — the NOT, AND and buffer cells add no noise in-model — and it inherits Part I's CW-figure caveats. The pre-amplified-variant comparison uses a fixed "CW − 1.9 dB" per-pulse calibration inherited from Part I; community review (July 2026) attributes most of that deficit to measurement convention and solver calibration (drive detuning, per-slot-peak metric, compression) rather than transient physics, with no basis for temperature independence — conservative for the pre-amplifier no-go, whose margin is orders larger, but flagged for recalibration. The promoted July 2026 notes now quantify that recalibration: with the parent's

proposed cavity-retuning fix applied in a variant run (the released chain still ships the zero-drift 582.80 nm length; Section 3.1), the per-pulse deficit is 0.44 dB on the retuned solver (continuous-wave 9.661 dB against the retuned solver's 9.225 dB; notes/2026-07-12-predictions-resolved.md), so the hardcoded 1.9 dB constant overstates it by ≈ 1.5 dB — still in the conservative direction; re-baselining the pre-amplified variant is a WP5 follow-up. The two elementary Gaussian rules are verified against exact Fock-space evolution to relative error < 10^{-4} at the points the package checks — loss+thermal Lindblad at coherent amplitude $\alpha = 3$, $\bar{n} = 2$, $\eta = 0.5$, and the two-mode-squeeze amplifier at $\alpha = 2$, $G = 2$ with a vacuum and a one-quantum thermal idler (9.9×10^{-5} on the thermal case) — and the assembled symbol-error expression is verified by direct Monte-Carlo sampling; the checks run as part of the package (qlindblad.py, run by run_all.py; qnoise.py). Neither check point is the operating one: certification at the operating occupation, gain and amplitude is open (notes/2026-08-05-idler-added-noise-and-the-nf4db-requirement.md §6 and limitation 2). The single-plasmon loss-dephasing threshold quoted in Section 3.3 is a standard consequence of that same verified generator but is printed by no released module; it is cited there as a hand derivation rather than a chain output, and turning it into a self-checked results.json key is a WP5 item. Kinetic's review adds a scope condition on that item: the $\exp[-(1-\eta)(2\bar{n}+1)N/2]$ factor is the leading-order-in-loss form, and at $\eta = 0.607$ the loss is not small, so the self-check should validate the finite-loss form before the N* values are quoted beyond order of magnitude.

Module	Role	Key outputs
qconstants.py	operating point + quantum scales	$\hbar\omega_0$, T_Q, $\bar{n}(T)$, $\tau(T)$ with 150-K saturation; Part-I anchors reproduced
qmode.py	mode quantization, Eqs. (Q1–Q3)	ε_1, knee/rail quanta, χ/κ, blockade size
qnoise.py	Gaussian calculus	loss/amp/combine rules; Eq.-(7) equivalence; symbol error + MC check
qdecode.py	flash decoder	2 comparators + NOT + AND + buffer; truth table b_1b_0 = A+B
qmac.py	QMAC-1	levels, 2-bit/1-bit errors, preamp no-go, weighted demo, bits/shot
qlindblad.py	verification	Fock-space checks of both Gaussian rules at fixed check points, < 10^{-4} relative
qerrors.py	temperature sweeps	Figure Q3/Table Q3 curves, threshold temperatures, floors
run_all.py / figures.py	driver / figures	results.json; Figures Q1–Q5

Table QA1. The released quantum model chain, module by module. Available at https://github.com/ryoji-info/FableComputer.

Appendix QB. Symbols

Symbol	Meaning	Value at the operating point
$\hbar\omega_0$	one plasmon at 1 THz	4.14 meV (6.63×10^{-22} J)

T_Q	thermal crossover $\hbar\omega_0/k_B$	48.0 K
$\bar{n}(T)$	thermal occupation of the mode	5.8 at 300 K; 1.2 at 77 K
ε_1	single-plasmon density amplitude, Eq. (Q2)	1.62×10^{-3}
N_knee, N_rail	quanta at the 1-dB knee / the coded rail	38 / 3833
χ, c_K	single-plasmon Kerr rate and its O(1) coefficient	$\chi/2\pi \approx 0.7$ MHz (c_K = 0.25, band 0.05–1)
κ	mode energy decay rate $1/\tau$	10^{12} s^{-1} at 300 K; 5×10^{11} s^{-1} saturated
x, V	measured quadrature and its variance	vacuum V = ½; coherent mean $\sqrt{(2N)}$
N_op	quanta per input at logic 1	400 (default; launch-budget-capped, a cap whose denomination is undefined — Section 5)
σ_rel, S	launcher amplitude jitter: pulse-to-pulse s.d. of the launched logic-1 amplitude over its mean, at the in-fabric launch plane; coded SNR S = $-20\log_{10}\sigma$_rel	10 % as coded (S = 20 dB); ≤ 8.16 % for the 2-bit error to reach 10^{-3} at some temperature (Section 4.2)
k	comparator threshold sharpness	≥ 16 (Part-II decoder requirement, no Part-I achievability basis; Part-I logic uses ≥ 8 as a design rule the released cell does not deliver in-model — caption, Section 5)
b_1b_0	decoded 2-bit word	= A+B in the digital case

Table QB1. Part-II nomenclature; Part-I symbols (s, L, M_th, Q, τ, F, G) carry over unchanged. Two entries carry caveats: N_op, whose launch-budget cap Section 5 qualifies, and k. The k ≥ 16 decoder rule is a Part-II requirement with no Part-I achievability basis behind it: Part I defines k against the cell's rail-to-rail input swing, which is ≈5× the ladder range this decoder works on, so a Part-I cell re-trimmed onto the ladder would present k_dec ≈ 1.6 (from Part I's k = 8 design rule) to ≈3.2 (from k = 16) — below Part I's own regeneration floor. In-model, k_dec = 16 therefore demands a comparator with ≈10× the β/Δn of a Part-I logic cell at that design rule: a redesign, not a re-trim, since trim moves the threshold n_th and is nowhere claimed — though nowhere explicitly excluded — to rescale β/Δn. The design rule itself is not delivered in-model: for a carrier-frequency input the released cell's only density coupling, rectification, gives 0.18–0.26 on Part-I Eq. (5)'s tanh scale at the delivered swing, and its amplitude transfer is concave, with no threshold (notes/2026-10-16-no-density-route-to-a-not-in-the-released-cell.md). Whether the ladder comparator can be a Part-I cell at all is open, and is the sharpest question this Part hands to the bench (notes/2026-07-15-finite-sharpness-is-not-a-variance.md §3, promoted 3-of-3).

Data and code availability

The quantum model chain is released as the runnable Python package fable-model-quantum alongside Part I's fable-model-chain; run_all.py reproduces every quantitative claim this manuscript takes from the package's results (results.json) and figures.py regenerates Figures Q1–Q5; values obtained by calling released functions at other stated arguments, or by exact algebra on them (Section 5's precision at N_op = 1916 and its 4.000-bit ceiling), reproduce from those calls; numbers quoted from the promoted community notes (notes/) reproduce from the listings and released-code calls those notes document (in the notes or their linked discussions), and those quoted from Part III from its listings. The package is archived with the manuscript at

the same repository as Part I, beside fable-model-ternary/, the listings of Part III [18]. The complete revision history of the manuscripts is kept in papers/REVISION-HISTORY.md; corrections adopted by the community record but not yet in a deposited revision are indexed in papers/ERRATA.md. The repository's open issues (https://github.com/ryoji-info/FableComputer/issues) are cited as open design questions, not as results (issue #140).

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